PAYLOAD BUS BREAKOUT

ARK Pixhawk Payload Bus Breakout

Pixhawk Payload Bus to Pixhawk-standard JST-GH, with a protected 5 V peripheral rail

The USA-built ARK Pixhawk Payload Bus Breakout fans the 30-pin Payload Bus out to individually keyed Pixhawk-standard JST-GH connectors. UART4/I2C3, CAN, Ethernet, USB, and GPIO each get their own connector.

A 5 V supply enters on a 6-pin Molex Pico-Clasp using the Pixhawk power-module pinout. An onboard TI BQ24315 protection switch derives the VDD_5V_PERIPH rail with a 1.5 A current limit, overvoltage protection, and a fault flag, and is gated by the autopilot over VDD_5V_PERIPH_nEN.

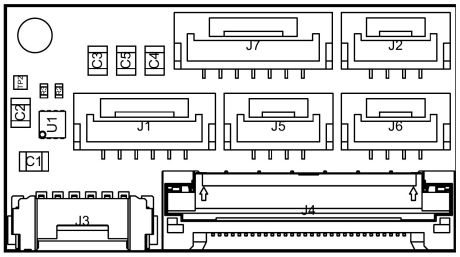
KEY FEATURES

- 30-pin, 0.5 mm pitch Payload Bus FFC connector
- UART4 and I2C3 on 6-pin JST-GH
- CAN on 4-pin JST-GH
- Ethernet on 4-pin JST-GH
- USB on 4-pin JST-GH
- GPIO, FMU PWM, and capture on 7-pin JST-GH
- 1.5 A protected 5 V peripheral rail (TI BQ24315): overvoltage, overcurrent, fault flag
- Includes 30-pin, 0.5 mm pitch, 150 mm FFC cable

SPECIFICATIONS

ELECTRICAL	
Supply voltage	5 V (J3)
Peripheral rail	VDD_5V_PERIPH, 1.5 A current limit
Rail enable	VDD_5V_PERIPH_nEN (active low)
Protection	Overvoltage, overcurrent, fault flag
Signal logic level	3.3 V
Operating temperature	−25 °C to +85 °C
Storage temperature	−25 °C to +85 °C

MECHANICAL	
Dimensions	34.25 × 18.50 × 5.81 mm
Weight	3.5 g



AT A GLANCE	
Payload bus	30-pin FFC
Interfaces	UART, I2C, CAN, Ethernet, USB, GPIO
Peripheral rail	5 V @ 1.5 A
Supply	5 V
Dimensions	34.25 × 18.5 × 5.81 mm
Weight	3.5 g
Hardware rev	Rev 1.0

COMPLIANCE & SOURCING

› Designed and manufactured in the USA

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-pixhawk-payload-bus-breakout/

DOCS docs.arkelectron.com/products/accessories/ark-pixhawk-payload-bus-breakout

CAD github.com/ARK-Electronics/ark-hardware

PAYLOAD BUS CONNECTOR

J4 — 30-pin, 0.5 mm pitch FFC (Amphenol F33M1B7F121030E200)

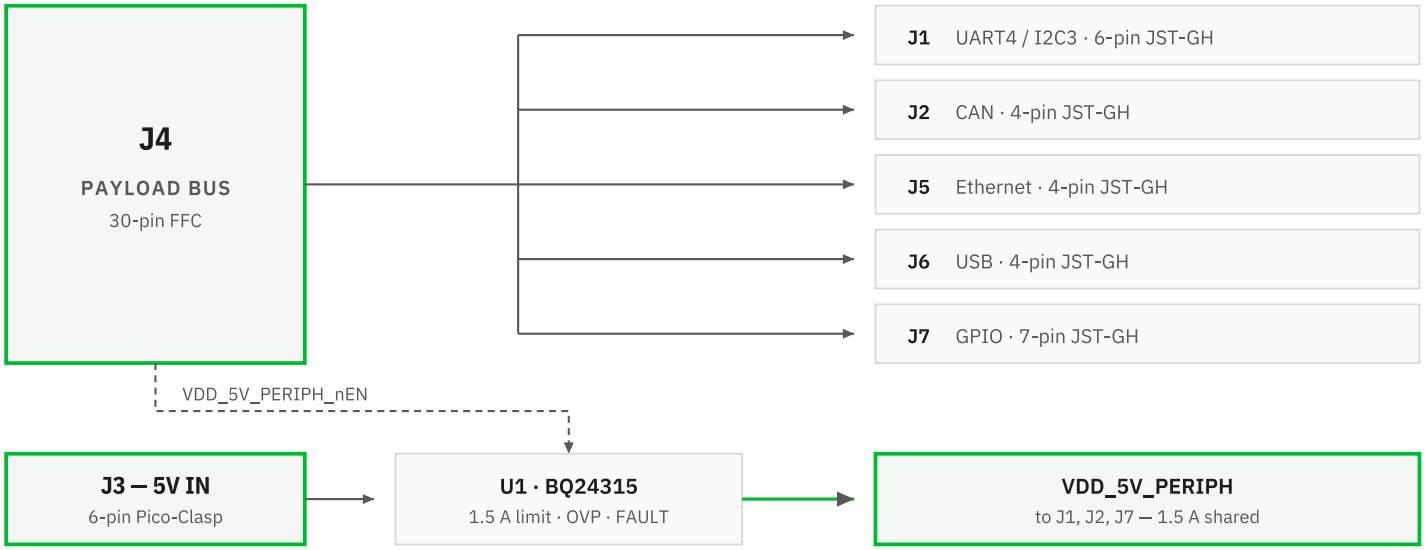
PIN	SIGNAL	LEVEL
1	GND	GND
2	UART4_TX_EXT	3.3 V
3	UART4_RX_EXT	3.3 V
4	GND	GND
5	I2C3_SDA_EXT	3.3 V
6	I2C3_SCL_EXT	3.3 V
7	GND	GND
8	CAN_P	5 V
9	CAN_N	5 V
10	GND	GND
11	FMU_CH7_EXT	3.3 V
12	GPIO12_EXT	3.3 V
13	FMU_CH8_EXT	3.3 V
14	GND	GND
15	FMU_CAP_EXT	3.3 V
16	GND	GND

PINS 17–30

PIN	SIGNAL	LEVEL
17	PYLD_ETH_TX_P	Diff.
18	PYLD_ETH_TX_N	Diff.
19	GND	GND
20	PYLD_ETH_RX_P	Diff.
21	PYLD_ETH_RX_N	Diff.
22	GND	GND
23	GPIO01	3.3 V
24	VBUS	5 V
25	USB_N	3.3 V
26	USB_P	3.3 V
27	GND	GND
28	VDD_5V_HIGHPOWER_nEN	3.3 V
29	VDD_5V_PERIPH_nEN	3.3 V
30	GND	GND
G1–G6	GND	GND

Pin 28 goes only to test point TP1. Pulling pin 29 low enables VDD_5V_PERIPH.

BLOCK DIAGRAM



BREAKOUT CONNECTORS

J3 — 5 V input, 6-pin Pico-Clasp

PIN	SIGNAL	LEVEL
1	5V	5 V
2	5V	5 V
3	I2C3_SCL_EXT	3.3 V
4	I2C3_SDA_EXT	3.3 V
5	GND	GND
6	GND	GND

J1 — UART4 / I2C3, 6-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5 V
2	UART4_TX_EXT	3.3 V
3	UART4_RX_EXT	3.3 V
4	I2C3_SCL_EXT	3.3 V
5	I2C3_SDA_EXT	3.3 V
6	GND	GND

J2 — CAN, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5 V
2	CAN_P	5 V
3	CAN_N	5 V
4	GND	GND

J5 — Ethernet, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	PYLD_ETH_RX_N	Diff.
2	PYLD_ETH_RX_P	Diff.
3	PYLD_ETH_TX_N	Diff.
4	PYLD_ETH_TX_P	Diff.

J6 — USB, 4-pin JST-GH

PIN	SIGNAL	LEVEL
1	VBUS	5 V
2	USB_N	3.3 V
3	USB_P	3.3 V
4	GND	GND

J7 — GPIO, 7-pin JST-GH

PIN	SIGNAL	LEVEL
1	VDD_5V_PERIPH	5 V
2	FMU_CH7_EXT	3.3 V
3	FMU_CH8_EXT	3.3 V
4	FMU_CAP_EXT	3.3 V
5	GPIO01	3.3 V
6	GPIO12_EXT	3.3 V
7	GND	GND

NOTES

- VDD_5V_PERIPH is shared by J1, J2, and J7 and is limited to 1.5 A in total.
- VBUS on J6 is the host's 5 V from Payload Bus pin 24, passed through unswitched and unprotected.
- J5 carries only the Ethernet pairs.
- Test points: TP1 = VDD_5V_HIGHPOWER_nEN, TP2 = VDD_5V_PERIPH, TP3 = FAULT (open drain).

ARK-DS-ARK-PIXHAWK-PAYLOAD-BUS-BREAKOUT | Version 1.0, 2026-10-29 | Compiled from the ARK Pixhawk Payload Bus Breakout Rev 1.0 schematic, the Rev 1.0 bill of materials, and the Rev 1.0 manufacturing documentation. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.